



SentimentMapper: A framework for mapping of sentiments towards disaster response using social media data

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Abstract

Social networking platforms have been generating a massive amount of data in real-time that can be analysed and used to support government and relief organizations in preparing quick and effective action plans for disaster response. Effective disaster response requires a broad understanding of disaster situations, such as the emergency necessities of the people, their sentiments towards emergency needs, and the geographical distribution of their requirements and opinions. However, in literature, many studies exist that estimate the emotions and sentiments of the people during a disaster; they are inept in identifying and mapping the public sentiments toward emergency needs. This paper proposes a framework called *SentimentMapper*. This framework quickly maps the sentiments of people toward emergency needs using social media data to plan for effective disaster response. In order to perform an automatic analysis of sentiments using Twitter (re-branded to X since July 2023) data, we introduce a BERT Convolutional Neural Network (BCNN). BCNN performs the sentiment analysis of the collected data from the disaster-affected people regarding essential needs like food, shelter, medical emergency, and rescue during different disasters. Next, we present a tweet-text independent approach to detect the location of the tweets posted on Twitter and discover the impacts in different areas due to any disaster event. Furthermore, we also study the variations in public attitudes about the essential needs during identical or different disasters. As a case study, the proposed framework has been used on the dataset collected from Twitter during the Assam flood 2021 in India and validated with the corresponding survey reports published by the government agency. The detailed results of the analytics in the proposed framework and its validation with the case study data confirm that it is capable of providing credible situational information quickly required for the disaster responses.

Keywords Natural disaster · Twitter · Sentiment analysis · Sentiment mapping · Geo-location · Emergency needs distribution

1 Introduction

During a disaster, victims and eyewitnesses communicate their feelings and opinion through social networking platforms like Twitter (re-branded to X since July 2023), Facebook, and others. Social communication platforms allow users to notify events and share their personal experiences and reactions immediately after a disaster event. Information like needs of the victims and their geographic locations

can strengthen the situational awareness, which improves the vision of the disaster response team. However, gaining a clear understanding of the situation can be difficult during an emergency due to its chaotic nature. Whereas, the real-time nature of the social sites can make them an effective tool for the disaster managers to manage the disaster situation adequately and promptly [70], which, in turn, will reduce the losses of lives and the economy. It is crucial to make important decisions quickly to help the disaster-affected people with their needs during the disaster response.

The widespread adoption of social communication platforms, especially Twitter, has emerged as a vital source of real-time information during crises [14]. Individuals often share critical updates, including geo-locations and urgent resource needs such as food, water, and medicine [66]. This timely exchange of information plays a pivotal role in facilitating effective emergency response efforts [59].

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The availability of precise and timely data can significantly empower disaster managers to formulate effective disaster response plans and provide timely assistance to the affected victims. Moreover, the rapid identification and distribution of sentiments expressed by the victims towards their emergency needs can play a pivotal role in enhancing resource allocation and aid planning by decision-makers. By harnessing real-time sentiment analysis, decision-makers can gain crucial insights into the emotions and requirements of those impacted by the disaster, enabling more efficient and targeted resource distribution for disaster relief efforts.

Although, in literature, many studies exist that estimate the emotions and sentiments of the people during disaster [47], they are inept in identifying and mapping the opinion of people toward emergency needs during a disaster. Natural disaster management is time crucial and thus demands prompt and adequate information gathering and analysis. Due to the lack of location information, the emergency responders cannot reach the affected person with appropriate aid, leading to poor disaster response. The visual representation of spatial dispersal of the public requirements and their attitudes towards relief supply during disasters is essential for the fast and efficient distribution of humanitarian needs. Even though the administration and concerned rescue organizations try to help the affected community after a disaster, people rarely acquire the full advantages as there is no adequate norm to locate and understand the exact needs. The primary objective of this research is to harness the wealth of information present on social communication platforms to enhance disaster response strategies and aid planning during emergencies. To achieve this, we aim to address existing gaps in the literature and develop a novel framework that geographically maps the sentiments of people on various emergency needs during different disasters. By bridging these gaps, our research endeavors to provide valuable insights into public sentiment, enabling more effective disaster response measures tailored to specific geographical areas and emergency requirements.

The main contributions of the proposed work are as follows.

- We analyze and categorize the various emergency requirements (e.g., food, shelter, rescue and medical emergency) of the people during and after the disaster.
- We develop an ensemble encoder called *Dictionary of Wisdom (DoW)* to fetch the semantic orientation from the text of the tweets.
- We propose a tweet text-independent framework to confine the location of the user. It allows decision-makers like relief agencies and rescue personnel to visually examine the spatial distribution of the catastrophic impact of the disaster and help them serve accordingly to the affected community.

- We devise the BCNN (BERT Convolutional Neural Network) model to classify the sentiments of the people regarding emergency supplies and compare its efficiency with both lexicon-based and machine learning-based models.
- We conduct this research from two perspectives to comprehensively understand public attitudes during the disasters. First, the sentiments of people during different disasters are visually examined, including their primary concerns and requirements for emergency relief supplies. On the other hand, it reveals how the views of the people towards different relief items changes in the case of identical disasters.
- Finally, we conducted a case study of the Assam flood 2021 in India to demonstrate the application of the proposed framework and its effectiveness.

We believe that both the sentiment analysis and the location information from the social media data during a disaster will help disaster managers to develop a robust and timely situational awareness of the disaster site.

The rest of the paper is organized as follows. Section 2 provides the relevant literature survey. Section 3 describes the motivation of this work. Section 4 presents the dataset details including natural disaster selection and data collection. The details of the proposed framework is discussed in Section 5 and the performance evolution with comparative analysis is presented in Section 6. Section 7 presents a case study on Assam flood 2021 in India. Section 8 discussed about the limitations of the proposed work. Finally, Section 9 concludes this paper.

2 Literature review

In this section, we discuss the literature review in two phases: the usage of social media data in disaster management and the role of the sentiment analysis of social media data for disaster risk reduction.

2.1 Role of social media in disaster management

In past decade, we saw a deluge in the use of social media platforms like Instagram, Facebook, Twitter, Flickr and LinkedIn for information sharing among social groups. Social media is a platform to express one's views and feelings in real-time. The disaster management cycle has four main phases called preparedness, response, recovery and mitigation. However, the social media plays a crucial role during three phases: preparedness (i.e., before the disaster), response (i.e., during the disaster), recovery (i.e., after the disaster). Before a disaster, it can assist people in preparing for a disaster and understanding which relief organizations will aid their communities. During

a disaster, it supports the users to communicate directly and share information with their relatives, friends, first responders, relief agencies, and humanitarian organizations. It also prevents the spreading of rumors because the organizations can quickly validate the facts. After the disaster, it helps to bring the community together to confer on the disaster event, to share the facts, experiences, and to coordinate response activities to mitigate the severe impacts of the disaster. The real-time feature of social media sites establishes them as an essential mean for disaster management, as both affected communities and disaster managers can post their queries, concerns and solutions in one place.

In the literature, many authors explored the benefits of utilizing social networking platforms and social media data for disaster management, especially for disaster risk reduction (DRR). For example, when a disaster occurs it is vital to deliver lucid situational information about the disaster site from the social media platforms to the organizations involved in crucial decision-making. Moreover, the analytics of social media data can enable the emergency responders and organizations to recognize and visualize the essential needs of the affected residents so that the relief resources and actions can be taken more efficiently [68]. However, our work focuses more narrowly on the usage of social media data during disaster response. In the literature, the social media data has been used in different ways for DRR such as situational awareness [12], information sharing and extraction [53], geographical information estimation [13], disaster damage assessment [48]. This paper explores the effectiveness of social media data for situational awareness, information sharing and geographical information estimation, which are crucial for adequate disaster response and DRR.

2.1.1 Situational awareness for DRR

Situational awareness identifies, processes, and comprehends the crucial elements of the disaster-hit areas and provides valuable insight into the situation. It informs disaster managers of what is surfacing in disaster situations and helps them to conceive a precise understanding of the impact of a disaster on the community. Precisely, the text classification and sentiment analysis through data analytic approaches are the primary tasks associated with the situational awareness.

2.1.2 Information sharing for DRR

Information sharing is how people express and communicate information, feelings or views via social networking platforms regarding various issues like natural disasters. Since social networking platforms offer a two-way communication channel for fetching and broadcasting information, disaster managers can predominantly use it for a quick and effective

disaster response [41]. In addition, this information sharing enables the emergency responders and relief organizations to assess the primary necessities of an affected community so that the relief supply and humanitarian actions are assigned efficiently. Both situational awareness and information sharing are used to expedite the disaster relief and mitigate the impacts of a natural disaster.

2.1.3 Geographical location inferences for DRR

Geographic information determination is the process of extracting location information to assess the widespread and devastating impacts of disasters over the disaster-hit areas. Location information is vital to comprehend the impacts of a disaster consisting of where the destruction happened, where the people require aid, and where these humanitarian aid are available. Moreover, extracting the locations from the disaster-related micro-blogs in social media data is essential for raising the situational awareness and speeding up the disaster response activities. The estimated location of the disaster victims helps the relief organization to map and track the disaster, which, in turn, improves the disaster response [56].

In literature, several techniques were reported to determine the location information from the social media posts, especially from the Twitter messages (i.e., tweets). Most of the existing work primarily employed the geotagged posts to determine the user location. [31] proposed a language model of locations to infer the tweet locations at the city level. Their model relies on the geotagged tweets to compute the tweet location. Many articles considered the geographic coordinates to be the most descriptive and accurate information for better disaster response and recovery [33]. However, many users do not explicitly specify their location due to privacy concerns and according to [9], tweets with geographic coordinates are rare (less than 1%). Moreover, Twitter also terminated the permission for precise geo-tagging on mobile devices. Therefore, the situational information of the disaster-hit area cannot be obtained entirely from geo-tagged tweets.

Several researchers widely used the historic post of the users to determine their geographical locations from their online posts, specifically from the tweets. For example, [35] proposed a global location recognition approach called Glitter that unites multiple micro-blogs of a user and uses them to recognize their locations. Another approach considered the associations between a location and its relevant keywords from the previous posts and inferring the location of a new post in a disaster scenario [25].

Other approaches confined the user location relying on the content of the tweet. Most of the current work follows the gazetteer-based methods [38] and the Named Entity Recognition (NER) methods [60] to extract the location information of Tweeter user from the Twitter messages. A gazetteer is a

geographical index of location names used to find the areas to which the location name is associated. Commonly used gazetteers include GeoNames¹, DB-pedia² and the USGS-database³. In this approach, the words in the Twitter message are peeked up in the gazetteer to discover the associated location. The gazetteer-based approaches do not require training data and are easy to execute, but have a comparatively low processing rate. Moreover, since users frequently use abbreviated names in tweets, these abbreviated phrases cannot be appropriately fetched using a location gazetteer. Therefore, the performances of these methods are poor for short and noisy text messages.

The second approach is NER that tokenizes each word of the text message using language-specific part-of-speech (PoS) tagging technique of natural language processing (NLP) [71]. It then recognizes the most likely group of words related to the name. However, this method works for well-written English sentences, not for tweets; because they contain shortened words, grammatical errors, and misspellings. In summary, gazetteer-based approaches and NER methods need a pre-defined set of attributes and site-specific gazetteers as input to extract the location information from the tweet. Thus, the performance of these methods highly depends on feature building.

2.2 Role of sentiment analysis in disaster response

Sentiment analysis is a data mining technique that deals with public sentiments in the social media posts using NLP, computational linguistics and text analysis. It determines the public feelings, attitudes, emotions, concerns and opinions towards a topic, which can be valuable information in different problems where decision making may depend on this analysis. Extracting the public sentiments from their messages posted on social networking platforms can contribute to the situational awareness by determining the polarity of sentiments expressed by the community [7, 37]. Following such sentiment patterns, decision-makers can determine the worst affected areas and needs of the people without additional efforts for information gathering. For example, [47] performed the sentiment analysis of the social media data (Twitter data) during the Hurricane Sandy, US in 2012. The sentiments of the users were visualized on a map centered around the hurricane and it revealed that the opinions of users depend on the location and the distance from the disaster-hit area. Ragini and Anand [52] proposed a sentiment analysis model, which gathers the disaster-related tweets and classifies them as per the requirements of the affected community. Furthermore, Dong et al. [11] explore the use of sentiment

analysis on social media data to enhance disaster response. The study employs eight machine learning models to classify public sentiment about relief efforts and essential needs. The findings reveal variations in public sentiment across disaster types and underscore the importance of sentiment analysis in identifying immediate needs. However, the study does not address the localization of sentiments, which limits its effectiveness in real-time disaster risk reduction.

Sentiment analysis is mainly concerned with determining the polarity of sentiments expressed by the people, who can express positive, negative, or neutral sentiments towards a subject or topic. Therefore, sentiment analysis is also termed as the polarity determination. There are two primary methods for conducting sentiment analysis: lexicon-based approaches and machine learning (ML) approaches. Lexicon-based approach utilizes a dictionary of pre-tagged words to analyze the polarity of a text and it assumes that the collective polarity of a sentence is the sum of the polarities of each individual word. In this approach, each word of a sentence is compared with the dictionary to get the polarity score and then these scores are summed up to get the overall polarity score of the sentence. The text is classified as positive, if the overall polarity score is positive; otherwise, it is negative. With the popularity of ML and big data analysis, researchers in the domain of disaster management adapted new techniques for better results [1, 30, 34]. ML-based methods are used to build a model by feature selection and learning from labeled training datasets. Supervised learning algorithms perform better than unsupervised learning algorithms [49], so this research relies on supervised ML-based approaches to execute the classification tasks.

Although, in literature, many studies exist those used the public emotions and sentiments during a disaster; they are inadequate in recognizing and localizing the feeling of the disaster victims towards the humanitarian aid they received. Moreover, the state-of-the-art techniques do not provide any general framework that can work for different disasters.

3 Motivation, problem statement and overview

3.1 Motivation

DRR for natural hazards is usually time crucial and requires prompt information gathering and analysis. An effective disaster response requires an exhaustive knowledge of disaster situations, for example, the geographical location of the victims, their requirements, and sentiments on different types of philanthropic aid they receive. Due to a lack of location information, the emergency responders cannot reach each and every affected person, which leads to poor disaster response. The precise locations of the affected people is very much

¹ <https://www.geonames.org/>

² <https://www.dbpedia.org/>

³ https://geonames.usgs.gov/domestic/download_data.html

needed to any organizations for the timely distribution of humanitarian aid during a disaster. Knowing the location and opinion of the impacted people regarding emergency needs will support the first responders, disaster managers and all concerned personnel in rapid situational awareness across time and space.

As we mentioned in Section 2.1, several approaches were proposed in the literature to determine the location of user from his social media post. However, most of the recent studies and techniques extracted the geographical location information of the victims from their geotagged social media post or by their post content using NER or gazetteer-based methods [62]. A study by [9] over a random sample of more than one million users discovered that less than 27% of Twitter users cited their location in their posts and only 0.42% of tweets were geotagged. Despite the advancement of information mining and geographical location computation techniques, credible and faithful automated mapping of the geographical locations of the victims during a disaster remains a challenge. Again, employing the existing techniques, the resolution of an evaluated location tends to be at the city or state level; obtaining more granular information like area or lane number is still challenging. These motivate us to develop a framework that can confine the location of the users of social networking platforms at a more granular level and independent from tweets posted by them.

Moreover, the existing studies and approaches for investigating the disaster scenarios generally focused on a particular disaster instead of presenting a general framework that works for different disasters. These research gaps motivate us to propose a generalized framework that can be used for different disasters. Furthermore, this framework will be a single automated platform, which can serve as a complete end-to-end system for analyzing and visualizing the geographical distribution of the sentiments of victims towards the emergency relief supply.

3.2 Problem statement

Given the Twitter text data as input, we need to conduct the sentiment analysis of the affected people toward various emergency needs. Furthermore, we aim to extract locations of the users from the Twitter data to gain insights of the geographical distribution of these sentiments about different types of emergencies.

3.3 Overview of the proposed work

In this paper, we proposed a tweet text-independent approach to extract the location information during a disaster event in a specific region. On contrary to the existing disaster mapping methods, we perform geo-parsing of almost all the tweets in real-time. We propose a generalized framework namely *SentimentMapper*, which can analyze the sentiments of the disaster-affected people towards different humanitarian aid like food or shelter, and map the geographical distribution of their needs and opinions. We believe that such a framework will facilitate the rescue personnel, disaster managers, and government authorities to deliver the targeted aid based on the geographical distribution of public sentiments during a disaster.

In this section, we provide an overview of the proposed work, which is depicted in Fig. 1. Section 5.1 discusses about collection of social media data (Twitter). In this work, we considered 12 disastrous events that occurred during 2018 to 2020 across the globe. Section 5.2 provides the detailed steps we used for data pre-processing and categorization after the raw data collection. Based on the requirements of the victims, we categorized the collected tweets into four categories, food, emergency medical services (EMS), shelter and rescue. Section 5.3 presents the details of the proposed framework for extracting the sentiments of people towards the humanitarian aid that they receive during and after the

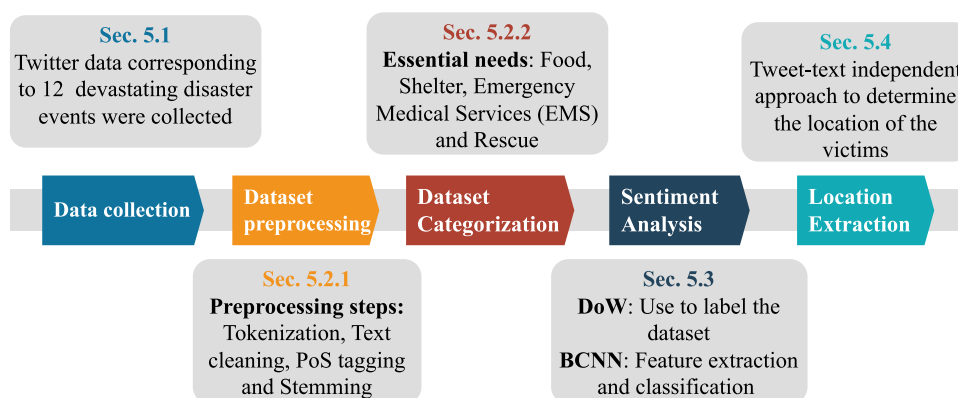


Fig. 1 Overview of the proposed work

disaster. Here we proposed two methods: Dictionary of Wisdom (DoW) for generating the true labels corresponding to the collected dataset, and BERT-Convolutional Neural Network (BCNN) for the sentiment analysis of the social media data. Section 5.4 presents a method that is independent of the tweet text to localize the Twitter post. Next, the performance evaluation of the proposed framework during the identical and different disasters is presented in Section 6. The application of proposed framework is discussed in Section 7 as a case study for the Assam floods in India in 2021. Finally, the limitations of the proposed work are discussed in Section 8.

4 Dataset details

While selecting the natural disasters, our objective to perform the in-depth analysis of public opinions during different and similar disasters by examining the sentiments of people of the disaster-hit area from social media data. In this work, we considered 12 destructive disaster events that occurred between 2018 to 2020 worldwide. These disaster events covered the four most devastating natural disasters: Hurricane, Flood, Cyclone, and Wildfire. The dataset includes three hurricane events namely, hurricane Florence (2018), hurricane Dorian (2019) and hurricane Eta (2020); three flood events namely, Kerala flood (2018), Midwestern U.S. flood (2019) and Hyderabad flood (2020); four cyclone events namely, cyclone Idai (2019), cyclone Nisarga (2020), cyclone Nivar (2020) and cyclone Amphan (2020); and two forest-fire events namely, California forest-fire (2018) and Australian forest-fire (2020). The details like event date, economic loss, and the number of affected people, corresponding to the aforementioned disasters collected from various news articles and reports are listed in Table 1.

5 Proposed framework: *SentimentMapper*

We propose a ML model, which evaluates the sentiments of people toward various identified needs and visualizes the geographical distribution of the disaster relief requirements using Twitter data. Figure 2 depicts the detailed block diagram of the proposed framework *SentimentMapper*, which consists of four modules. The first module M_1 is responsible for the data collection and the second module M_2 is designed to pre-process and categorize the input data. The sentiment analysis module M_3 is developed to analyze the sentiments of victims towards humanitarian aids during a disaster. The last module M_4 is devised to extract the tweet locations and visualize the geographical distribution of the sentiments of victims about the emergency needs. The proposed framework can provide a micro-level understanding to the disaster managers regarding the crisis and assist them in preparing a more efficient and effective disaster response plan. Next, we present the detailed descriptions of each module one by one.

5.1 Data collection

In Fig. 2, module M_1 give the brief overview of the data collection module of the proposed framework. Twitter is a popular micro-blog and social network platform, on which users can interact and share their thoughts through messages known as “tweets”. Twitter has more than 100 million vibrant users, who post approximately 340 million messages daily. During a disaster, the sufferers can post crucial disaster-related information and their fear, like the distribution and severity of damage caused by the disaster. Therefore, Twitter is an essential social networking platform for effective and quick disaster response, as it allows rapid access and provides the updates of critical information shared by the common people. Therefore, in this work, Twitter is considered for the

Table 1 Details of Natural Disasters

Sl.no.	Disaster	Year	Location	Event date	Economic loss	Affected people
1	Hurricane Florence [57]	2018	Florence, Italy	Aug 31, 2018	\$24.23 billion	> 1 million
2	California wildfires [51]	2018	Western U.S	Nov 08, 2018	\$25.4 billion	183
3	Kerala floods [17]	2018	South India	Aug 16, 2018	\$56 billion	> 1 million
4	Cyclone Idai [5]	2019	Beira city, Mozambique	Mar 14, 2019	≥ \$2.2 billion	> 8.7 million
5	Hurricane Dorian [40]	2019	Bahamas	Sep 06, 2019	\$3.4 billion	> 70k
6	Midwestern US floods [2]	2019	Midwestern U.S	Mar 11, 2019	\$2.9 billion	14 million
7	Cyclone Nisarga [65]	2020	India	June 01, 2020	\$803 million	> 10k
8	Cyclone Nivar [46]	2020	South India, Sri Lanka	Nov 23, 2020	\$600 million	> 10k
9	Hyderabad floods [22]	2020	South, East India	Oct 11, 2020	\$681 million	> 4k
10	Cyclone Amphan [58]	2020	East India, Bangladesh	May 16, 2020	\$13.7 billion	> 50k
11	Australian wildfire [54]	2020	Australia	Jan 01, 2020	\$103 billion	> 7k
12	Hurricane Eta [6]	2020	Central America	Nov 03, 2020	\$8.3 billion	> 50k

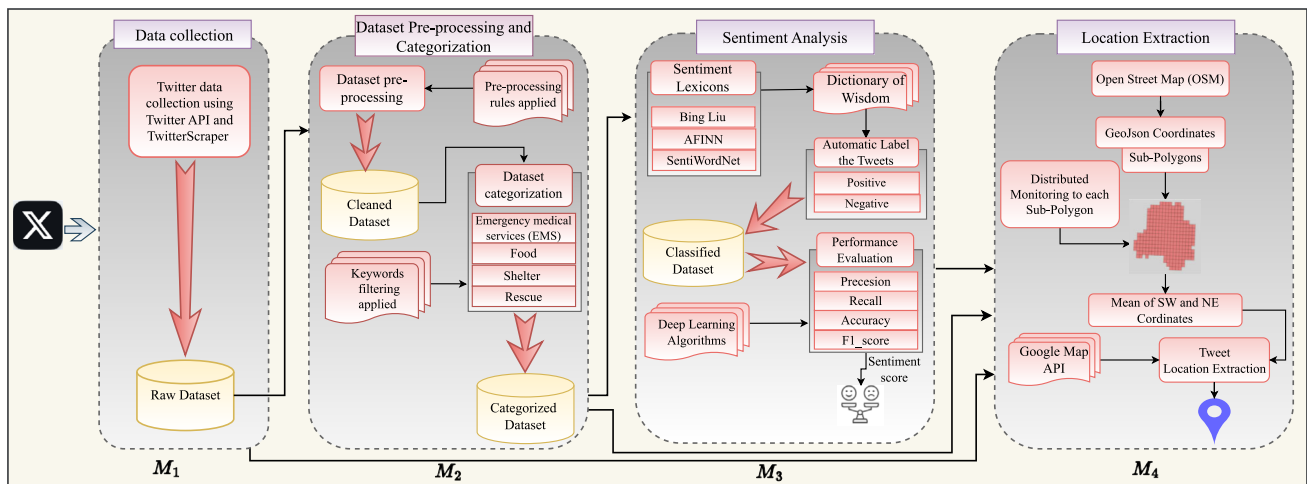


Fig. 2 Detailed overview of the proposed framework

collection of social media data and we create the Twitter dataset comprised of 38,923 tweets for the sentiment analysis. A part of these tweets was collected using a Python-based library Tweepy [55]. It is an open-source python package that makes it convenient to use Twitter API with its own sets of classes and methods. Twitter allows access to only the last seven days' tweets using its streaming API, and there is no provision to gather the historical data. Hence, the researchers employ the TwitterScraper API in Python for the rest of the data gathering [63]. Examples of explored keywords used for data gathering are "cyclonenisarga", "nisarga", "cyclone", "nisargacyclone" and "cyclone Maharashtra India". The percentage dispersion of tweets among diverse disasters are depicted in the pie-chart shown in Fig. 3. Further, Fig. 4 represents the word cloud created from the tweets belonging to four different categories of emergency needs in the compiled Tweeter dataset. A word cloud is a pictorial representation of the most frequently used words in a given dataset. The font size of the word indicates the frequency of the word in the respective corpus. That means the words with larger font sizes are more common than those with the smaller font sizes. The word cloud can help the emergency responders to get an overview of the sentiments towards the particular category of humanitarian aids. In the case of the word cloud of our compiled dataset, the most frequently occurring positive words are 'help', 'food', 'relief', 'God', 'thank', 'save', 'heal', 'clean', and many more. On the other hand, the most frequently occurring negative words are 'lack', 'failure', 'deadly', 'victim', 'terrible', 'need', 'destroy' and many more. The words like 'relief', 'save' and 'heal' represent the positive attitudes of the affected people towards the supply of humanitarian aids. The copious number of words like 'failure', 'terrible', 'dark', 'delay', 'killed' and 'deadly' indicate that by such tweets people seek immediate helps from the emergency responders.

5.2 Dataset pre-processing and categorization

This module is designed to filter the input data and make it usable for ML algorithms used in the next module of the proposed model. Module M_2 in Fig. 2 depicts the flow of the dataset pre-processing and categorization.

5.2.1 Dataset pre-processing

Twitter is the most popular platform for micro-blogging that allows people to send brief updates through micro-blogs or 'tweets'. The length of the 'tweets' are limited to 280 characters; so abbreviations and informal phrasing are common in a tweet. Due to character restrictions, tweets contain limited information, so people are more likely to use shortened words in tweets. The primary purpose of this step is to pre-process the disaster data and make it usable for the ML algorithms in the next stage to pull the features accurately. The pre-processing phase comprises of four steps namely 'tokenization', 'text cleaning', 'part-of-speech

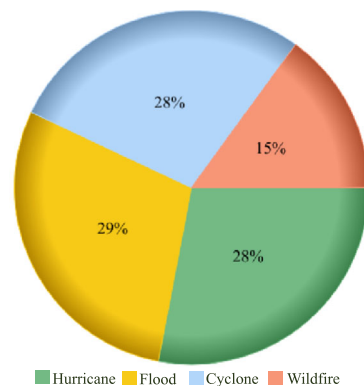


Fig. 3 The percentage distribution of tweets in various natural disasters



Fig. 4 Word cloud obtained from the compiled dataset for four different categories of emergency needs

(PoS) tagging' and 'stemming'. In the first step, it starts with tokenizing the input text into separate terms known as "tokens". Twitter messages contain links, tabs, newlines, foreign language words, symbols, numerals, repeated words and emoticons. The text cleaning step removes the insignificant textual data from the tweets by removing non-ASCII characters, symbols, links, tabs, hashtag signs, and others. In the third step called PoS tagging, we label each token of text corpus with its appropriate part of speech. The final step of text pre-processing is stemming, which is a process used to extract the original form of the words by removing inflection. For example, the word "eats" is transformed into the stem word "eat". Figure 5 shows the output of each data pre-processing step with an example tweet. Here *NN*, *VB*, *ADJ*, and *NUM* corresponds to noun, verb, adjective and number, respectively.

5.2.2 Dataset categorization

In this work, we categorize the various essential requirements of the people (i.e., humanitarian aids) during and after the disaster into four categories namely food, shelter, EMS and rescue. The percentage distribution of the tweets regarding the humanitarian aids during different disasters is shown in Fig. 6. The keyword filtering technique was applied to categorize the disaster related tweets among these categories [67]. A vast collection of keywords is built considering the keywords of multiple research papers [11, 52, 66, 69]. Each keyword has a significance to the associated category. Table 2 depicts the coined keywords associated with each of these four categories and Table 3 shows the distribution of tweets among various categories. Efficient data categorization eventually

helps the disaster manager in developing faith, strength, and belief among victims during the disasters.

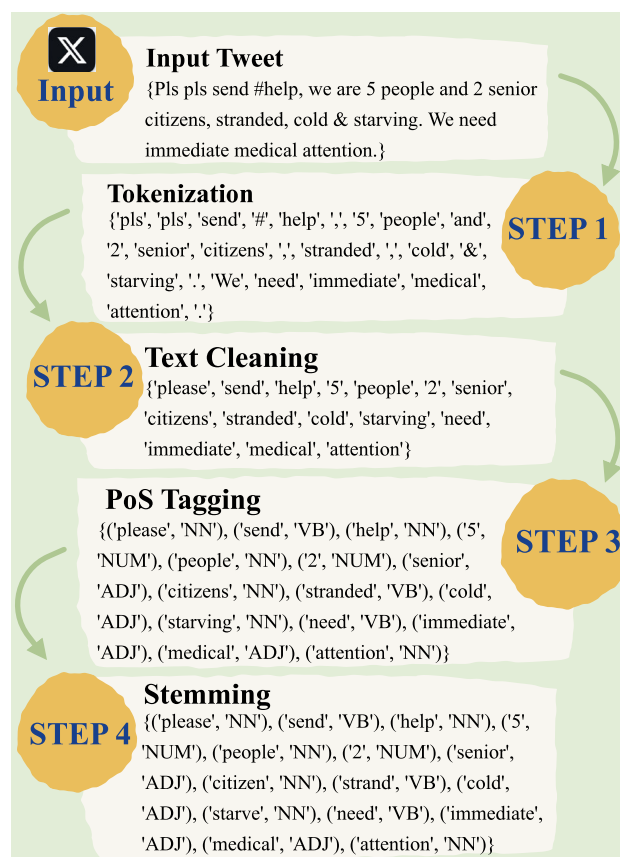


Fig. 5 Detailed block diagram with an example tweet for data pre-processing

Table 2 List of considered categories and associated keywords

Category	Keywords
Food	<i>food, starve, hungry, milk, drinking water, thirsty, dehydrated, eat, clean water, thirst, hygienic food</i>
Shelter	<i>shelter, living place, accommodation, rest, safe place, clean area, sleep, house, living area</i>
EMS	<i>medicine, hospital, doctor, first aid, casualties, injured, injury, fatalities, people dead, people died, collapsed, urgent, destroyed, damaged, missing, tragic, tragedy, critical, affected, victims</i>
Rescue	<i>trapped, stranded, help, stuck, save, danger, caught, critical, evacuate, survivor, rescue, hazard, risk</i>

5.3 Sentiment analysis module

This module is comprised of two components (a) automatic labeling of the Twitter data, and (b) extracting the sentiments of people towards the humanitarian aids they received during and after the disaster. The first component is responsible for automatically labeling the Twitter dataset to each category using the *DoW*. The second component relies on BERT (i.e., Bidirectional Encoder Representations from Transformers) and CNN (i.e., Convolutional Neural Network) to classify the Tweeter texts based on the sentiments of the people during and after the disaster. Module M_3 in Fig. 2 gives a brief overview of the sentiment analysis module of the proposed framework. The detailed descriptions these two component are elaborated below.

5.3.1 Dictionary of wisdom (DoW)

As the raw dataset is compiled by employing an open-source Python library called Tweepy running on the top

of twitter rest APIs, the compiled dataset is not labeled. A labeled dataset is required before the model training step and there exists several approaches to generate such labels. In the semi-supervised learning approach, it uses previously human-labeled tweets to train the model and then it labels the rest of the unlabeled dataset considering them as true labels. On the other hand, the lexicon-based approach can also be used to label the datasets. The difference in these two approaches is that the results of the semi-supervised approach could be inconsistent, whereas the lexicon-based models can give a better accuracy [45]. Therefore, the results of these models are more widely accepted.

In this work, we utilize the lexicon-based method to automate the process of computing the semantic orientation of words present in the text. These lexicons are the dictionary of words that contribute to identifying the polarity of the text. The foremost benefit of considering a lexicon-based method is that it can be easily altered and understood by a human being. The semantic orientation of the text can be captured and tagged as neutral, positive, or negative using this method. There are two types of sentences: subjective and objective (or neutral). Subjective sentences are those that have sentiment-related information and segregation of such sentences is important to analyze the needs of the victims. However, objective (neutral) sentences are the ones that have no information of sentiments and these sentences will not help the emergency responder to analyze the need of the people. Therefore, in this work, we considered only the positive and negative sentiments of the people and discard the neutral sentiments.

Bing Liu [39], AFINN encoder [10], SentiStrength [64], SenticNet [8] and Sentiwordnet [4] are some of the existing benchmark models using as lexicon-based dictionaries for sentiment analysis. In order to classify the dataset, the

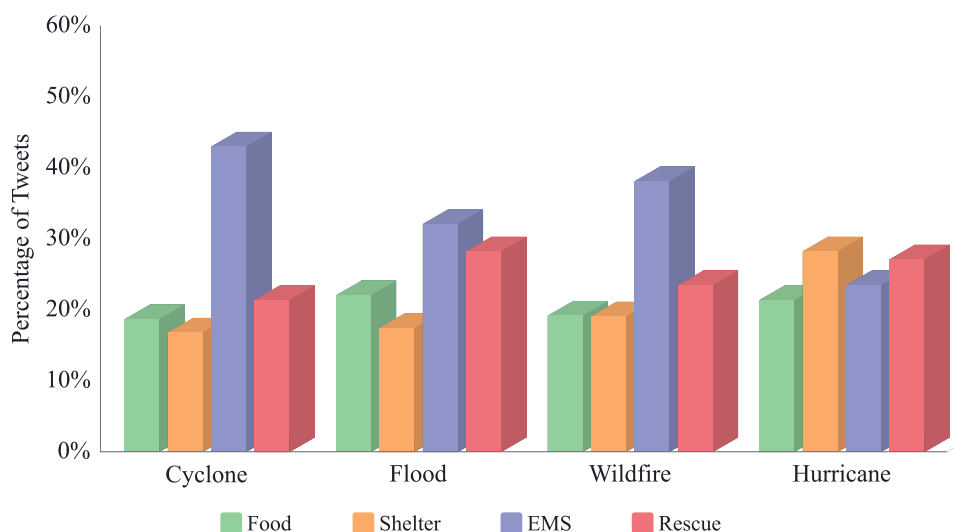


Fig. 6 The percentage distribution of tweets regarding different essential needs during different natural disasters

Table 3 Distribution of the disaster related tweets in various categories

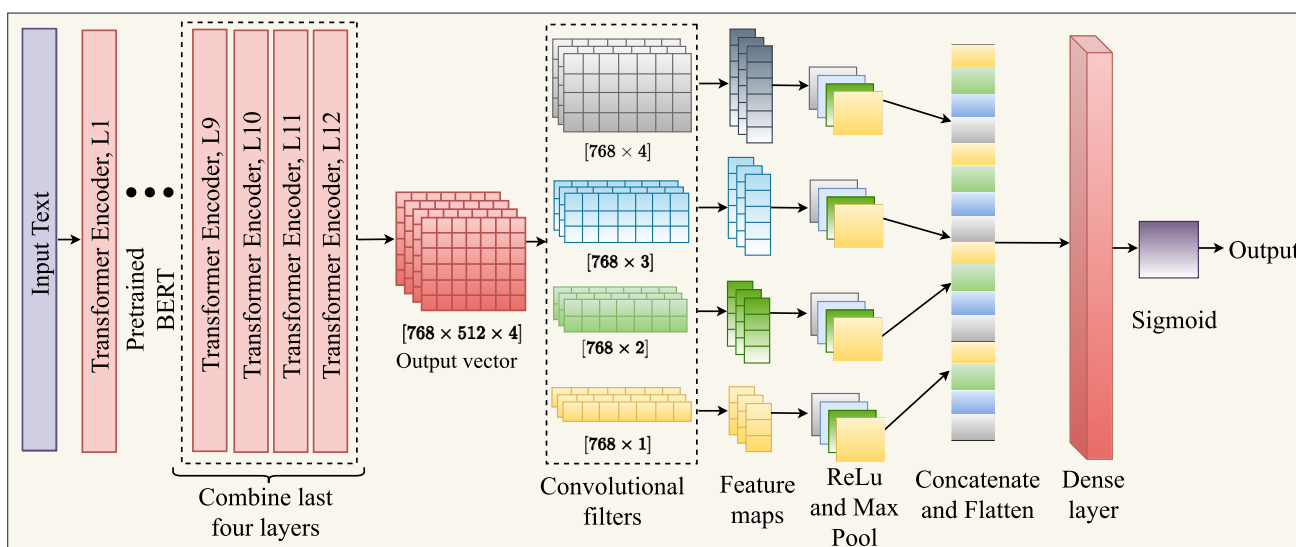
S.no.	Disaster	Year	People's requirements				Tweets per disaster
			Food	Shelter	EMS	Rescue	
1	Hurricane Florence	2018	671	949	727	1293	3640
2	California wildfires	2018	294	1543	329	1035	3201
3	Kerala floods	2018	991	1840	554	1594	4979
4	Cyclone Idai	2019	470	1637	323	819	3249
5	Hurricane Dorian	2019	727	1064	734	1095	3620
6	Midwestern US floods	2019	816	1032	760	1024	3632
7	Cyclone Nisarga	2020	543	930	562	569	2604
8	Cyclone Nivar	2020	558	781	530	597	2466
9	Hyderabad floods	2020	623	778	591	673	2665
10	Cyclone Amphan	2020	290	1129	356	724	2499
11	Australian wildfire	2020	373	1052	314	700	2439
12	Hurricane Eta	2020	804	1291	778	1056	3929
Total no. of Tweets							38,923

proposed framework utilizes three publicly available sentiment lexicons namely Bing Liu, AFINN and Sentiwordnet. Instead of using just one model to label the tweets, we combined the predictions of all three models to reduce the error rate and increase the classification accuracy. Moreover, the assumed lexicons comprises the complete list of slang and misspelled words those are pertinent for the classification of Tweet texts. Furthermore, combination of these three models gives us an ensemble encoder with the ‘wisdom of crowd’ minimizing the error rate to absolute zero. The ‘Wisdom of crowd’ refers to the thought that the aggregated decision of a group of individuals is more accurate than a single expert decision [32]. The key idea is that the perspective of an individual can inherently be biased, whereas incorporating the

aggregated knowledge of a group can eradicate the bias or noise and yield a more precise and intelligible result. This means better estimates are constructed by aggregating latent knowledge than by believing in the outcome of the single approach. Since the three approaches are based around the dictionary of words, these three are compiled as ‘*Dictionary of Wisdom*’.

5.3.2 BERT-Convolutional Neural Network (BCNN)

Social media data possess several vital information like the primary relief supplies that the victims lack, and their fulfillment and anxiety. It is essential to categorize the posts based on the sentiments of the people on the supplies of

**Fig. 7** The architecture of the proposed BCNN model

various humanitarian aids to capture the situational information and the needs of the affected areas. Several existing studies proposed various classifiers for the classification purpose by utilizing the deep learning (DL) techniques like Recurrent Neural Networks (RNNs) [19], and Long Short-term Memory (LSTM) [23]. Despite the advancements and implementation of these DL techniques, input embedding is a pivotal factor that degrades the performance of these approaches. Existing studies rely on the pre-trained word embedding packages to perceive textual details retrieved from the text messages. Since nearly all these packages are unidirectional and have fixed parameters, they may not perform well enough in different domains without fine-tuning their parameters. Kinsella *et al.* [28] proposed a state-of-the-art language model called BERT to improve the language representation of the text messages. This model learns the deep-bidirectional representations in all the layers by simultaneously training both left and right contexts. It embeds the token sequence, segment, and position in the final input feature vector of a given sentence (token). This feature allows the model to learn the meaning of sentences from all their contexts. For various textual tasks it can be fine-tuned or used directly as a feature extractor.

In this study, we propose a BCNN model that utilizes the contextual knowledge embedded in the pre-trained BERT model. Specifically, the model integrates embeddings derived from the last four hidden layers of BERT into the convolutional layers of a CNN, enhancing its ability to capture semantic and contextual features. The proposed model consists of two core components: (1) a BERT-based embedding generator and (2) a CNN-based feature extractor. The BERT base model, comprising 12 transformer layers, excels at generating rich, contextualized vector representations. Each transformer layer is characterized by a hidden size of 768, 12 self-attention heads, and a total of 110 million parameters, enabling the model to effectively capture intricate semantic relationships within the input text. For a given input sequence, BERT can process up to 512 tokens, producing embeddings with dimensions $e \in \mathbb{R}^{768 \times 512}$, which serve as the foundation for subsequent feature extraction and analysis. To further enhance the semantic richness of these representations, the outputs of the last four hidden layers of BERT are fused by concatenation to create a final 3D feature tensor, represented as E :

$$E = \{e_{-4}; e_{-3}; e_{-2}; e_{-1}\} \in \mathbb{R}^{768 \times 512 \times 4} \quad (1)$$

where e_{-k} denotes the output embedding from the k -th last hidden layer. This tensor E is then passed through four parallel convolutional layers with filters of sizes 768×1 , 768×2 , 768×3 , and 768×4 . These filter configurations are specifically tailored to capture uni-gram, bi-gram, tri-gram, and tetra-gram features, respectively, thereby extracting patterns

of varying granularity from the input representations. The convolution operation for a filter of size n can be mathematically expressed as:

$$f_{i,j}^n = \sigma \left(\sum_{k=1}^{768} \sum_{l=1}^n W_{k,l}^n \cdot E_{i+k-1,j+l-1} + b_n \right), \quad (2)$$

where W^n denotes the weight matrix of the convolutional filter with dimensions $768 \times n$, b_n represents the bias term, σ is the ReLU activation function applied element-wise and $f_{i,j}^n$ is the resulting feature map at position (i, j) .

After the convolutional operation, a global max-pooling layer is applied to each feature map to extract the most salient features. The pooling operation, as described in (3), mathematically selects the maximum value from each feature map generated by the convolutional layers.

$$p_n = \max \left(f_{i,j}^n \right), \quad (3)$$

where p_n represents the pooled output corresponding to the feature map derived from the convolutional filter of size n . The pooled features from the convolutional layers are concatenated to form a unified feature vector, as represented in (4).

$$F = \{p_1, p_2, p_3, p_4\}, \quad (4)$$

where F aggregates the most significant features extracted by the filters of varying sizes. This concatenated vector F is then flattened and passed through a dense layer, followed by a sigmoid activation function for binary classification. Let W_d and b_d denote the weight matrix and bias term of the dense layer, respectively. The output of the dense layer is computed using (5).

$$h = \sigma(W_d \cdot F + b_d), \quad (5)$$

where σ is the activation function (e.g., ReLU). Finally, the sigmoid function transforms h into a probability score for the positive sentiment class, as expressed in (6).

$$\hat{y} = \text{sigmoid}(h) = \frac{1}{1 + e^{-h}}, \quad (6)$$

The model was optimized using the binary cross-entropy loss function, mathematically defined in (7).

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N (y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)), \quad (7)$$

where N denotes the total number of training samples, y_i represents the true binary label for the i -th sample, and

$\hat{y}_i$ is the predicted probability for the corresponding sample. Figure 7 provides a detailed block diagram of the proposed BCNN model.

5.4 Location extraction module

Understanding how disaster situations evolve in different locations is vital for prioritizing its response efforts. During the disaster, victims and eyewitnesses use Twitter to spread awareness, share their feelings, opinion and convey information. Messages communicated by the eyewitnesses are helpful to collect on-the-ground facts about the disaster site; but most of the time the knowledge in these tweets is generic. For example, in case of the tweet like “If you’re from India-Kerala state. Calicut city- near this area, please help! am near them but we have flood please share!”, although the purpose of the message is understandable to the emergency responders, the access to the street-level location information is still challenging. The most commonly used approach to find out the locations from the tweets includes parts-of-speech (POS) tags. In this approach, the text documents can be parsed to extract likely geographic tokens or “named entities” (e.g., state or region such as “Kerala”). In other cases, where locations are mentioned in the tweets, geo-encoder can be used to retrieve the location information by looking up the location names on a map and returning the geo-coordinates.

Unfortunately, the tweets containing address information are significantly low and mostly mention only the name of states, cities, or districts [43]. To address this limitation, the M_4 module proposed a method that is independent of the tweet text to localize the Twitter post. It enables the unfolding of the disaster events corresponding to the specific region. Brief overview of the module M_4 is represented in Fig. 2.

Our approach differs from the existing tweets mapping methods as the proposed framework can geo-parse almost all tweets in real-time. This module employs a systematic

methodology to pinpoint the approximate geographic origins of tweets, down to the street or building level, while maintaining user privacy and adhering to Twitter’s Terms of Service. The proposed methodology is illustrated in Fig. 8, showcasing the operation of the Location Extraction Module. The process begins by extracting the GeoJSON polygon map of the desired city using OpenStreetMap (OSM) [20]. This polygon represents the geographical boundaries of the area under consideration and serves as the basis for further mapping. In order to simplify the subsequent computations, the latitude and longitude of the polygon’s extremities (north, south, east, and west) are identified. These coordinates are then used to create a bounding rectangular envelope, referred to as the ‘envelope’. The generated envelope acts as a simplified boundary encapsulating the irregular shape of the city’s GeoJSON polygon. Further, to systematically monitor the region of interest, a grid with dimensions $m \times n$ is created within the bounding envelope using square sub-polygons denoted as R_i , each having a diagonal length of d . This grid forms a mesh-like structure that overlays the envelope polygon. Once the grid is generated, it is overlaid on the GeoJSON polygon representing the city boundaries. The intersection of the grid with the city’s polygon is then evaluated. Among the total N sub-polygons ($N = m \times n$) in the grid, only those sub-polygons R_i that have a non-zero overlapping area with the GeoJSON polygon of the city are retained. These retained sub-polygons represent the parts of the grid that fall within the actual geographic boundaries of the city, ensuring that only the relevant regions are monitored. The total region to be monitored, denoted as (R_M) , can then be expressed as the union of the retained sub-polygons. This relationship is mathematically represented as (8):

$$R_M = \bigcup_{i=1}^N R_i = R_1 \cup R_2 \cdots \cup R_N \quad (8)$$

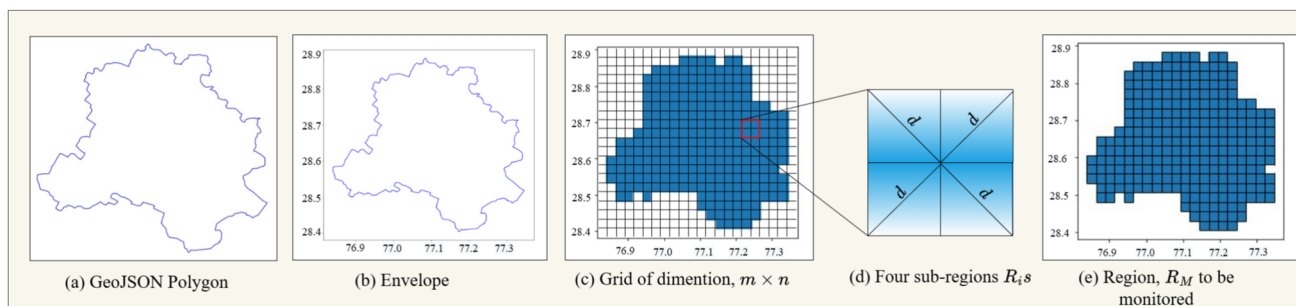


Fig. 8 An illustration of sub-division monitoring, (a) GeoJSON polygon of a region, (b) Corresponding axis-aligned envelope, (c) Grid of square sub-polygons of dimension $m \times n$, (d) Four square sub-regions (R_i s), and (e) Complete region, R_M to be monitored

where each sub-region R_i is defined as the intersection of its corresponding grid area a_i with the city's polygon area a_{cp} :

$$R_i = a_i \cap a_{cp} \quad (9)$$

here:

- a_{cp} represents the area of the city's polygon as obtained from the GeoJSON data.
- a_i represents the area of the i^{th} sub-polygon or sub-region R_i .

By applying this grid-based approach, the irregular polygon of the city is effectively divided into smaller, more uniform square sub-regions. This process simplifies the task of monitoring by converting the city's irregular geographic shape into a structured and slightly regular format. Each of these smaller sub-regions can then be analyzed individually, making the overall monitoring effort more efficient and scalable.

For instance, tweets received from a specific sub-region R_i are guaranteed to originate from the bounding area defined by the south-west (SW) and north-east (NE) latitude and longitude coordinates of that sub-region. The center of the SW-NE diagonal is designated as the approximate latitude and longitude for all collected tweets from R_i . These geolocation points are then plotted on a map using OSM, allowing responders to pinpoint the origin of tweets. This method enhances precision, enabling location tracking to reach down to the street or building level, rather than being confined to the broader city level. The module is designed with flexibility, allowing the size of the square sub-regions to be adjusted based on the desired resolution of mapping. However, selecting an optimal size is critical. For instance:

- If the squares are too small (e.g., 0.1 km^2), neighboring points may become indistinguishable on the map, complicating analysis. Additionally, this would result in an excessive number of sub-regions, leading to increased computational time and resource usage.
- Conversely, excessively large squares may generalize the location information, reducing accuracy and introducing ambiguity.

For effective disaster monitoring, a balance between computational efficiency and mapping precision is necessary. Based on extensive trials, the optimal diagonal length for square sub-regions is fixed at $d = 2 \text{ km}$. This value provides a suitable trade-off, ensuring high accuracy without significantly increasing computational complexity. However, for broader regions, such as a country, the square size can be increased to accommodate larger-scale monitoring requirements.

Earlier, the localization and estimation of the disaster impact on the community depend either on the satellite image processing or on the survey reports prepared by the government/non-government agencies. Furthermore, these reports and satellite images ignore the perspective of the disaster victims. The proposed model works in real-time by monitoring the tweets posted during and after the disaster. Once the data is gathered for any disaster event, the proposed framework performs the sentiment analysis and determines the victim's location that will assist the emergency responders to plan for response and rescue operations.

Social media data is heavily used to analyze and evaluate situations in times of disaster and derive decisions for the actions to be taken. This information can be used to support disaster managers in decision-making during emergency situations. During critical situations, it is not surprising that privacy is often seen as a secondary concern [18, 61]. Although, we consider the tweets related to people seeking help during disasters, we believe that privacy issues become a concern when personal information is disseminated indiscriminately. However, the proposed framework is not violating any privacy issues.

It is also necessary to give proper importance to the local values and people's emotion about their specific regions, which may be under protected wild life area. The proposed system is not going to do any analysis (location or sentiment) using tweets posted by the people during the period when there is no disaster. Rather, it starts collecting the tweets only when a disaster hits that area and this system will be used by the governmental or authorized organizations, who provides the post-disaster immediate responses. It is needless to say that individuals who share their location on social media during emergency situations often do so in the hopes of being rescued or located. The proposed system aimed to shed light on the ways in which social media can be used as a tool for rescue and support during emergencies. We believe that in such situations, where individuals may be in imminent danger, the sharing of information such as location can play a crucial role in ensuring their safety and well-being. When there is no disaster in an area, the proposed system is not going to collect the data (tweets by the people) to perform any analysis (location or sentiment) of the people living their normal lives in the protected wild life area. Hence, it will not violate any ethical and cultural values of that specific region.

6 Performance evaluation of SentimentMapper

The proposed model is evaluated using compiled disaster-related Twitter dataset. As mentioned in Section 4, the compiled dataset covers the four more devastating natural disasters namely hurricane, flood, cyclone and wildfire. The

experiments were conducted on Intel(R) Xeon(R) Gold 5118 processor with 96GB RAM having Quadro P2000 NVIDIA graphics card. All the methods are implemented and tested using the Python based ML libraries called tweepy, matplotlib, scikit-learn, NLTK and Google geocoding APIs. This section presents the prediction, visualization and analysis of the results obtained for performance evaluation of the proposed framework.

6.1 Sentiment analysis during different type of disasters

This section examines the public sentiments during four different types of disasters namely flood, wildfire, hurricane and cyclone. Figure 9 depicts the percentage distributions of the positive and negative sentiments of people during four different disasters and the distribution of their opinion towards four different relief supplies (food, shelter, EMS and rescue). Here we categorize the tweets regarding different essential needs using *DoW* to explore the public sentiments during those four disasters. It is observed that although the nature of each disaster is disparate, the patterns of people's sentiments about

different essential relief supplies during various disasters are similar. For example, Fig. 9 illustrates that when a disaster strikes, people have the most negative sentiments about EMS in case of any disaster. This implies that the people are most concerned about EMS during any disaster compared to other essential needs. It can also be noted that people have varying levels of essential needs when the disasters occur. Thus, the humanitarian organizations need to prepare pertinent plans to deliver the critical relief supplies during any disaster. In addition, Fig. 9(i)-(iv)(b) present the percentage distributions of people's sentiments (positive or negative) about the four different relief supplies. This complete picture can help the emergency responders and relief organizations to visualize the varied concerns of the people and to plan their response activities accordingly.

6.2 Sentiment analysis during identical type of disasters

This section analyses the public sentiments during identical types of disasters. Here we consider four different types of natural disasters namely flood, wildfire, hurricane and



Fig. 9 Public attitudes towards the humanitarian aids during four different natural disasters (flood, wildfire, hurricane and cyclone) categorized by *DoW*. (i)-(iv)(a) Percentage distributions of positive and negative

sentiments in case of four different natural disasters. (i)-(iv)(b) Percentage distributions of people sentiments regarding four different essential relief supplies

cyclone. The dataset includes three floods, two wildfires, three hurricanes and four cyclone events. Figure 10 depicts the distributions of tweets among the same type of disasters, whereas Fig. 11 presents the distributions of the positive and negative sentiments among those same type of disasters occurred in different locations at different times. Although many factors such as disaster intensity, the nature of the disaster, and community susceptibility influenced the positive and negative sentiments of the people about a disaster, we observed that the negative sentiments are higher than the positive sentiments during the cyclone events. It can also be noted that the proportion of the negative sentiments (indicating emotional distresses) among the people has increased over the years during the hurricanes (Fig. 11(c)).

Figure 12 compares the positive sentiments of the people towards the humanitarian aids during the same type of disasters occurred in different locations at different times. Such kinds of visual sentiment representations can help the disaster responders to understand the needs of the impacted people and to prepare accordingly to serve the affected community.

Here, we can observe that the people have varying opinions towards the emergency relief supply even for the same type of disasters. For example, people were more concerned about EMS during the California wildfire, whereas food was the primary concern of the people during the Australian wildfire. Furthermore, we can also observe that except for the Australian wildfire, EMS is the emergency relief supply that people were most concerned during other wildfire events. Thus, we conclude that the feelings of the people towards the essential needs can vary despite the fact that they are facing the same type of disasters. Hence, disaster relief organizations should prepare a suitable disaster response plan for all types of disasters.

6.3 Performance of BCNN model

This section examines the effectiveness of the proposed BCNN model in classifying positive and negative sentiments expressed by affected individuals. The precision, recall, $F1$ score and accuracy are computed to compare the performance

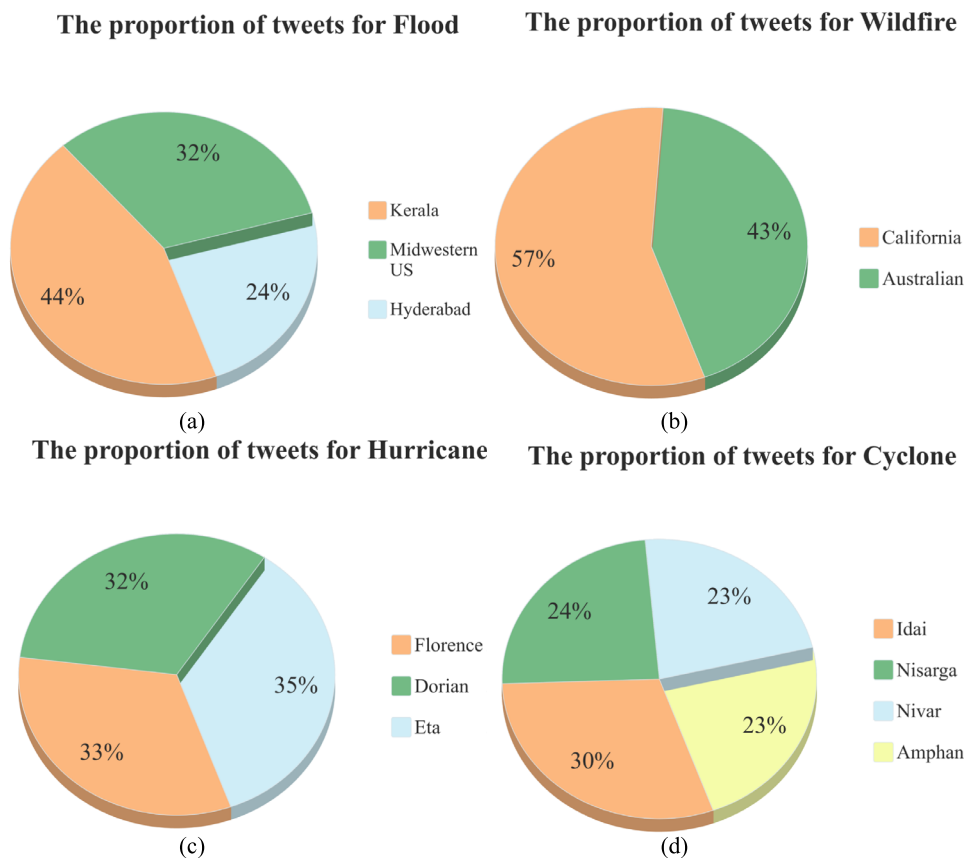


Fig. 10 Proportions of tweets for identical type of disasters like (a) Flood, (b) Wildfire, (c) Hurricane, and (d) Cyclone, occurred in different locations

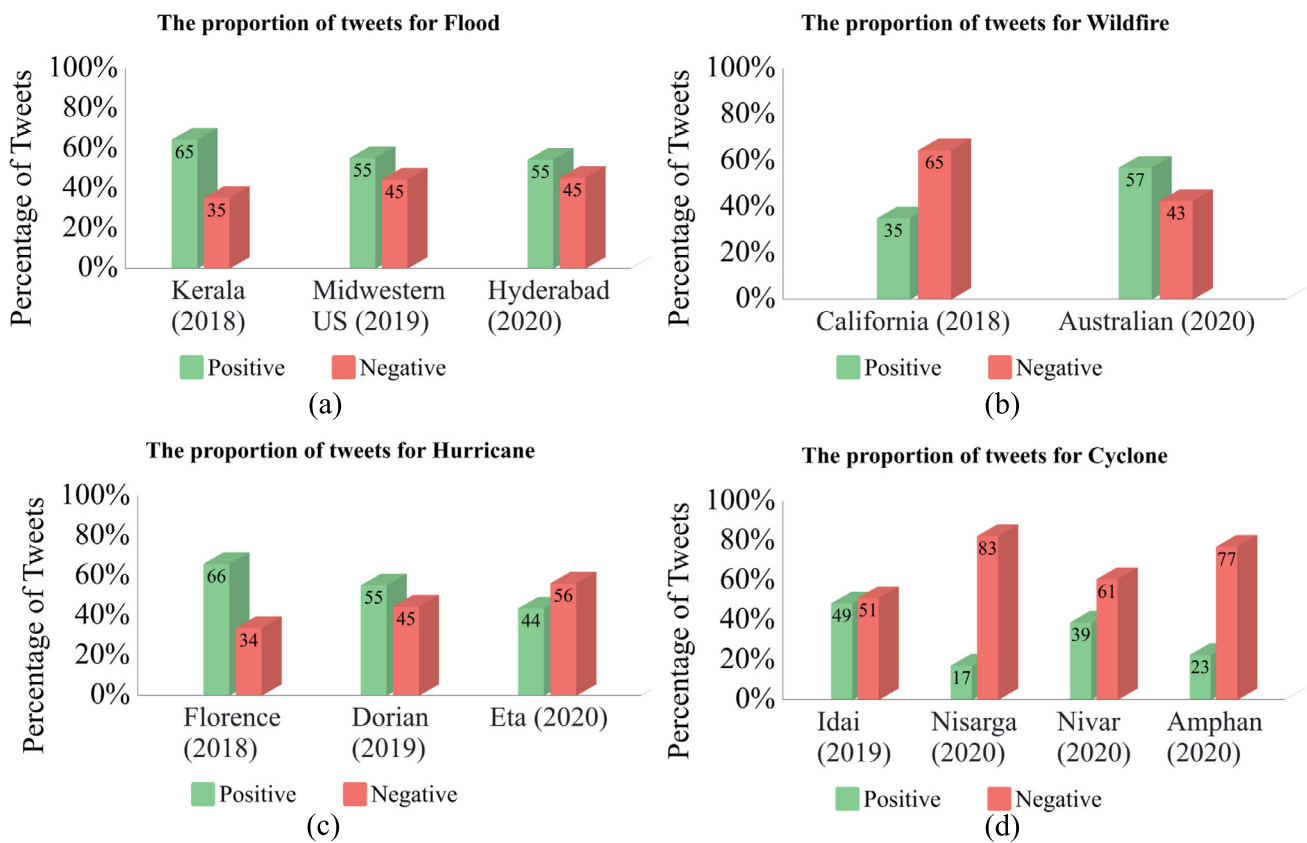


Fig. 11 Percentage distributions of the positive and negative tweets for identical type of disasters like (a) Flood, (b) Wildfire, (c) Hurricane, and (d) Cyclone, occurred in different locations

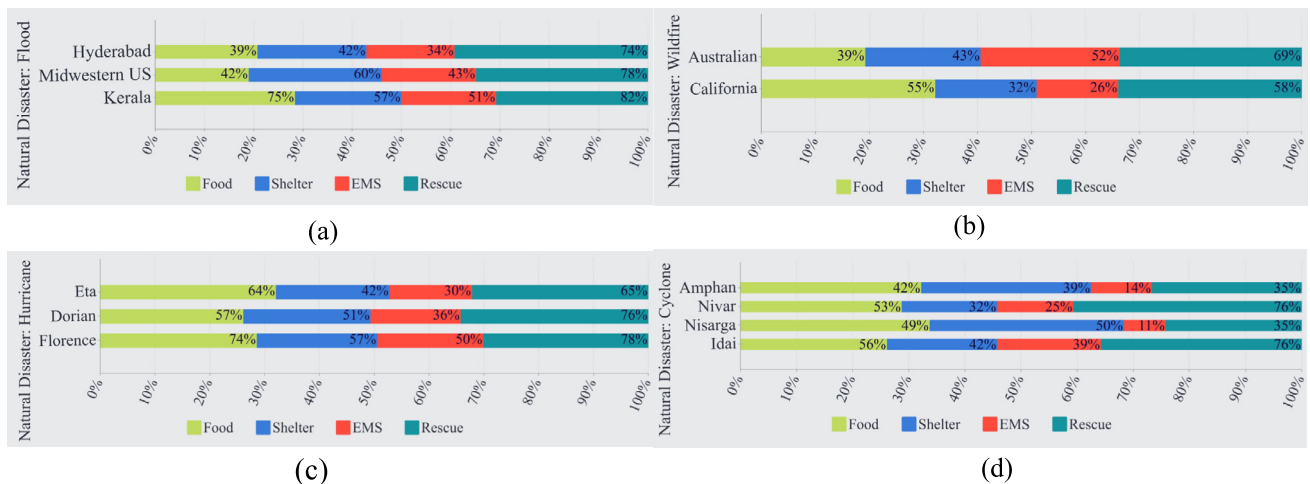


Fig. 12 Proportion of positive sentiments of people towards the supply of humanitarian aids during the identical type of disasters like (a) Flood, (b) Wildfire, (c) Hurricane, and (d) Cyclone, occurred in different locations

of the considered models using the following (10).

$$\begin{aligned}
 Precision &= \frac{TP}{TP + FP} \\
 Recall &= \frac{TP}{TP + FN} \\
 F1 &= 2 \times \frac{Precision \times Recall}{Precision + Recall} \\
 Accuracy &= \frac{TP + TN}{TP + TN + FP + FN}
 \end{aligned} \quad (10)$$

where TP denotes True Positives, TN means True Negatives, FP is False Positives, and FN presents False Negatives. These are the primary metrics for measuring the performance of a classification system. The precision is the ratio of correctly identified positive observations (TP) to the total number of identified positive observations ($TP + FP$). Whereas, the recall is the ratio of correctly identified positive observations (TP) to the total number positive observations ($TP + FN$) in the class. The $F1$ Score is the harmonic mean of precision and recall, and the accuracy is computed by dividing the correctly identified observations by the total observations.

6.3.1 Experimental setup

The experimental implementation was conducted on a GPU-accelerated computing environment utilizing an NVIDIA Quadro P2000 system. The experiments were performed on a machine running the Windows operating system, equipped with an Intel(R) Xeon(R) Gold 5118 processor and 96 GB of RAM, ensuring efficient computational performance. The models were developed using the Keras deep learning framework, with TensorFlow serving as the backend.

6.3.2 Hyperparameters tuning

The hyperparameters utilized in this study for the BCNN model are summarized in Table 4. The model was optimized using the Adam optimization algorithm with an initial learning rate of 0.001 to facilitate efficient gradient updates. A batch size of 8 was employed, and the model was trained for 17 epochs to ensure adequate learning without excessive computational overhead. To prevent overfitting and

Table 4 Set of hyperparameters identified for training model

Hyper-parameters	Training epochs	Batch size	Learning rate	Loss function	Dropout	Optimizer
Values	17	8	0.001	Binary cross-entropy	0.2	Adam

enhance generalization, an early stopping mechanism with a patience value of 3 was incorporated, allowing training to terminate when no significant improvement was observed. Additionally, a dropout rate of 0.20 was applied to enhance generalization. The model was trained using the binary cross-entropy loss function, which is well-suited for binary classification tasks.

6.3.3 Experimental results

In order to evaluate and validate the model, we split the dataset into two subsets: 70% for training and 30% for testing. We employ a five-fold cross-validation approach on the training set to ensure that the model does not over-fit and performs reliably on new, unseen data.

In the initial phase of the experiment, traditional machine learning algorithms are employed to evaluate their effectiveness in sentiment classification. Word2Vec embeddings are utilized as feature representations, which are subsequently used as input to eight classifiers: K-Nearest Neighbor (KNN), Support Vector Machine (SVM), Naive Bayes (NB), AdaBoost (AB), Decision Tree (DT), TreeBagger (TB), Logistic Regression (LR), and Random Forest (RF). The performance of these classifiers in identifying sentiments expressed by individuals during disasters is summarized in Table 5. The RF model achieved the highest metrics, with precision of 80.01%, recall of 79.71%, $F1$ score of 79.86%, and accuracy of 79.65%. Following closely, LR attained an $F1$ score of 75.60% and an accuracy of 76.41%. Ensemble methods such as AB and TB demonstrated moderate performance, with $F1$ scores of 66.43% and 68.39%, respectively, surpassing the simpler DT which yielded the lowest $F1$ score at 63.28%. NB and KNN exhibited limited effectiveness, presenting $F1$ scores of 64.14% and 69.60%, respectively. These results provide a baseline for comparison, offering insights into the capabilities of traditional machine learning methods in sentiment analysis within the disaster context.

Table 5 Performance evaluation of traditional classifiers for sentiment classification

Classifier	Precision	Recall	F1	Accuracy
KNN	65.24	74.58	69.60	71.21
SVM	66.14	69.57	67.81	67.12
NB	65.12	63.18	64.14	68.23
AB	67.13	65.74	66.43	69.81
DT	62.23	64.36	63.28	63.76
TB	66.74	70.12	68.39	70.23
LR	78.21	73.14	75.60	76.41
RF	80.01	79.71	79.86	79.65

Subsequently, the proposed BCNN model is benchmarked against the existing deep learning models, including TextCNN-Attention [37], LSTM [23], Bidirectional LSTM (BiLSTM) [29], CNN-BiLSTM [50], BERT [28], and LeBERT [44] to further validate its effectiveness in sentiment classification tasks. Table 6 presents a comparison of the performance of the proposed BCNN model against existing models. The proposed BCNN model achieved the highest precision (93.90%), recall (91.70%), F1 score (92.80%), and accuracy (93.45%), outperforming all benchmark models. LeBERT and CNN-BiLSTM followed closely, with F1 scores of 91.65% and 91.36% and accuracies of 93.10% and 91.10%, respectively, demonstrating their capability in sentiment classification. Other models, including BERT, BiLSTM, LSTM, and TextCNN-Attention, showed F1 scores ranging from 82.81% to 85.84%, reflecting a consistent but lower performance. The consistent performance of BCNN underscores its adaptability and effectiveness in classifying the sentiments of the people.

The effectiveness of the proposed BCNN framework is further evaluated by examining its capability to classify the sentiments of affected communities regarding various humanitarian aids. As shown in Table 6, the LeBERT model exhibits the best performance among the alternatives following BCNN. Consequently, in this experimental setup, the proposed BCNN framework is compared with both the LeBERT [44], a deep learning-based model, and the lexicon-based VADER model [24]. For this setup, the entire dataset of tweets is divided into training and testing sets in a 7:3 ratio for each of the four categories of humanitarian aids (food, shelter, EMS and rescue) across each disaster. The comparative results are presented in Tables 7, 8, 9, and 10, which illustrate the performance of the BCNN model alongside the LeBERT and VADER models for four distinct disaster types: cyclone, flood, wildfire, and hurricane. The results highlight the superior performance of the proposed BCNN model for sentiment analysis across various disaster types, including floods, wild-

fires, and hurricanes, as well as humanitarian aid categories such as food, shelter, EMS, and rescue. Compared to the LeBERT and VADER models, BCNN consistently achieves higher precision, recall, F1 score, and accuracy, demonstrating its ability to identify sentiments with greater accuracy and reliability. Notably, BCNN excels in precision, reaching up to 93% for cyclone-related rescue and 89% for wildfire-related EMS, indicating its effectiveness in minimizing false positives. The model also exhibits a strong recall, achieving 89% for cyclone-related Food and 85% for hurricane-related rescue, ensuring comprehensive sentiment coverage. Its balanced performance is reflected in the highest F1 scores across all categories, with scores of 91% for cyclone-related rescue and 86% for wildfire-related EMS. Furthermore, BCNN maintains consistently high accuracy values, such as 94% for cyclone-related Food, EMS, and Rescue, as well as 94% for hurricane-related EMS. This confirms its robustness in disaster-specific sentiment analysis.

Furthermore, to assess the robustness of the proposed framework in real-time scenarios, we utilized the Natural Hazards Twitter Dataset (NHTD) [11, 42], which captures sentiments expressed during natural disasters. By integrating this diverse dataset, we ensure a thorough evaluation of the framework's performance across various real-world contexts, thereby enhancing its comprehensiveness and validity. Table 11 presents a comparison of classification accuracy on the NHTD dataset, showcasing the performance of the BCNN model in relation to the Naive Bayes model, which was identified as the most effective approach in the study by Dong et al. [11]. The comparison underscores the significant advantage of the BCNN model, which consistently outperforms the Naive Bayes approach across all disaster scenarios. The BCNN not only demonstrates its ability to generalize effectively across diverse disaster types but also maintains consistently high accuracy, while the Naive Bayes model exhibits minor inconsistencies. For example, the Naive Bayes model achieves relatively lower accuracy for Hurricane Harvey (0.90) and Hurricane Dorian (0.95), whereas the BCNN maintains superior performance with accuracies of 0.96 and 0.98, respectively. Moreover, the BCNN achieves perfect accuracy (1.00) for both blizzards and wildfires, demonstrating its ability to handle even the most complex and nuanced scenarios with outstanding precision.

The consistent performance of BCNN can be attributed to its capacity to leverage contextual embeddings through BERT and effectively capture hierarchical features using its convolutional layers. These advantages empower BCNN to better grasp the nuanced language and sentiments found in disaster-related tweets, establishing it as a strong framework for disaster sentiment analysis and resource planning.

Table 6 Performance comparison of the proposed BCNN model with existing models

Model	Precision	Recall	F1	Accuracy
TextCNN-Attention [37]	83.72	81.91	82.81	82.35
LSTM [23]	81.73	83.92	82.81	81.76
BiLSTM [29]	82.14	84.13	83.12	84.87
CNN-BiLSTM [50]	91.70	91.03	91.36	91.10
BERT [28]	85.10	86.6	85.84	87.23
LeBERT [44]	92.00	91.30	91.65	93.10
Proposed (BCNN)	93.90	91.70	92.80	93.45

Table 7 Performance evaluation of the proposed BCNN model compared to LeBERT [44] and VADER [24] models for Cyclone-related humanitarian aid categories

Humanitarian aids Model	Food			Shelter			EMS			Rescue		
	LeBERT	VADER	BCNN	LeBERT	VADER	BCNN	LeBERT	VADER	BCNN	LeBERT	VADER	BCNN
Precision (%)	81	74	86	86	71	90	87	76	91	88	82	93
Recall (%)	86	77	89	79	73	83	84	74	87	86	83	89
F1 (%)	83	76	88	82	72	86	86	75	89	87	83	91
Accuracy (%)	92	89	94	90	81	93	93	78	94	89	86	94

Table 8 Performance evaluation of the proposed BCNN model compared to LeBERT [44] and VADER [24] models for Flood-related humanitarian aid categories

Humanitarian aids Model	Food			Shelter			EMS			Rescue		
	LeBERT	VADER	BCNN	LeBERT	VADER	BCNN	LeBERT	VADER	BCNN	LeBERT	VADER	BCNN
Precision (%)	82	78	84	77	73	81	83	77	85	85	83	87
Recall (%)	84	76	84	76	73	77	80	75	83	81	77	82
F1 (%)	83	77	84	77	73	79	82	76	84	83	80	84
Accuracy (%)	87	88	89	88	85	90	91	84	94	89	85	92

Table 9 Performance evaluation of the proposed BCNN model compared to LeBERT [44] and VADER [24] models for Wildfire-related humanitarian aid categories

Humanitarian aids Model	Food			Shelter			EMS			Rescue		
	LeBERT	VADER	BCNN	LeBERT	VADER	BCNN	LeBERT	VADER	BCNN	LeBERT	VADER	BCNN
Precision (%)	77	75	79	76	65	80	83	79	89	81	76	85
Recall (%)	78	74	78	76	73	77	84	81	84	80	72	83
F1 (%)	78	75	79	76	69	79	84	80	86	81	74	84
Accuracy (%)	87	86	89	85	76	91	92	85	93	84	83	86

Table 10 Performance evaluation of the proposed BCNN model compared to LeBERT [44] and VADER [24] models for Hurricane-related humanitarian aid categories

Humanitarian aids Model	Food			Shelter			EMS			Rescue		
	LeBERT	VADER	BCNN	LeBERT	VADER	BCNN	LeBERT	VADER	BCNN	LeBERT	VADER	BCNN
Precision (%)	76	75	79	78	72	78	81	75	86	81	76	83
Recall (%)	77	72	76	75	73	76	79	72	82	84	70	85
F1 (%)	77	74	78	77	73	77	80	74	84	83	73	84
Accuracy (%)	88	82	91	86	85	89	91	75	94	90	79	93

Table 11 Accuracy comparison of the proposed BCNN framework on the NHTD

Disaster	Dong et al. [11]	BCNN (Ours)
Tornado	0.94	0.98
Hurricane Sandy	0.97	0.99
Floods	0.97	0.98
Blizzard	0.98	1.00
Matthew	0.95	0.97
Hurricane Harvey	0.90	0.96
Hurricane Michael	0.98	0.99
Wildfires	0.99	1.00
Hurricane Dorian	0.95	0.98
Average	0.96	0.98

7 A case study: Assam flood 2021 in India

To demonstrate the application of the proposed framework *SentimentMapper*, we choose the Marigaon district of the Assam state of India as a study area. In August 2021, the situation in the north-eastern states of Assam worsened when a torrential rainstorm hit the area. According to the Assam State Disaster Management Authority (ASDMA) officials, more than 647 thousands of people were affected by the floods across 732 villages in 16 districts of Assam [3]. Marigaon, Nagaon, Darrang, Kamrup and Nalbari were some of the districts of Assam that were the worst hit during the Assam 2021 floods. Among these, Marigaon was the most susceptible to the flood inundation [27]. Marigaon district constitutes five revenue blocks (A – E): Marigaon, Mayong, Bhuragaon, Laharighat and Mikirbhetta.

Figure 13(a) depicts the bounding box over the study area, and Fig. 13(b) represents the local revenue sub-divisions (A – E) of the Marigaon district. As mentioned in Section 5.1, we utilized Tweepy for tweet-gathering purposes, resulting in

a total collection of 19,893 tweets between August 27 to September 5, 2021. These tweets were posted by individuals within the predefined bounding box that covers the study area.

After that, we filtered the flood-related tweets posted by the flood-affected people using the keywords “Assam”, “flood” and “Assam flood”. Subsequently, we left with 9,453 flood-relevant tweets for the selected duration. Furthermore, a grid of dimension 80×70 of square sub-polygons having a diagonal of 2 km was formed. After the intersection of the grid with the GeoJSON polygon of the study area, we left with 260 square sub-regions. Next, each sub-regions are monitored individually. The geographical distribution of the tweets based on the emergency needs (food, shelter, EMS and rescue) of the people is depicted in Fig. 13(c). The dispersal of tweets over the disaster-hit area represents the extent of disaster impacts across the different locations. The visual representation of the needs of victims assists the disaster managers in scrutinizing and sustaining the required commodities to serve the needy in the disaster-affected area. It is vital to understand the expansion of the disaster impacts to prioritize and schedule the disaster response efforts.

In this case study, we used the *SentimentMapper* to map the tweets, emergency needs, and sentiments of the people towards different humanitarian needs during a flood. Figure 14 represents the spatial distribution of tweet hotspots for different emergency needs over the Marigaon district. As shown in the figure, most of the rescue and EMS requests came from the South-West region (i.e., revenue block B) of the Marigaon district. This indicates that the impact of that flood in the South-West region is comparatively more extensive than the impacts in the other areas. The outcome also reveals that the people living near the Brahmaputra river banks were initially most affected by the water-logging. However, the rising water forced the residents to move away from the banks and looks for shelters. Although the food

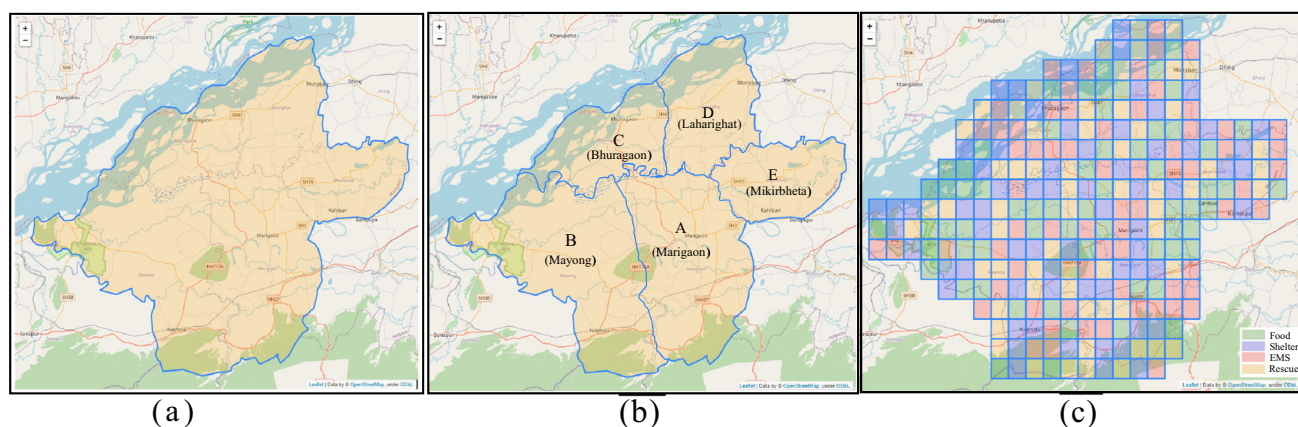


Fig. 13 (a) Bounding box over the Marigaon District, Assam, India. (b) Local revenue sub-divisions of the Marigaon district. (c) Spatial distribution of relevant tweets based on the different emergency needs like Food, Shelter, EMS, and Rescue

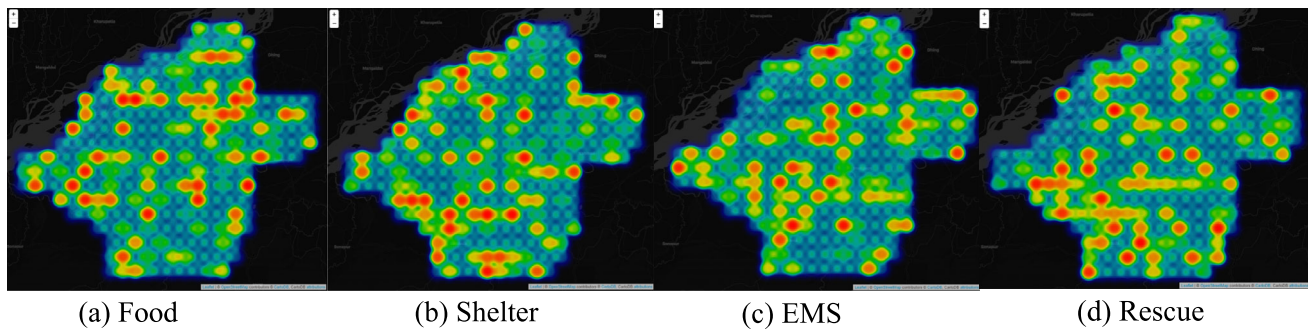


Fig. 14 Spatial distribution of tweet hotspots for different emergency needs, (a) Food, (b) Shelter, (c) EMS, and (d) Rescue

necessity was widespread throughout the region, the high demand for the food was in the South-West and North-East (i.e., revenue blocks B and D) regions of the Marigaon district. This is probably because the victims first migrated to the safest area and then searched for the food. The geographical distributions of positive and negative sentiments of the people toward four different emergency needs in each sub-region are presented in Fig. 15. The visual sentiment distribution helps the disaster managers bring the demanded materials to the specified location instead of dumping them where there is an excess supply. Moreover, the results assist the emergency responders and rescue personnel in reading the minds of impacted people and preparing accordingly to mitigate the impacts of the disaster on the community.

In order to investigate the trustworthiness of the situational information determined by the social networking platform using *SentimentMapper*, we validate the related information with the available published news articles and state government reports [15, 26]. For example, according to the cumulative flood report of ASDMA [3], the area near the Brahmaputra river of the Marigaon district was heavily inundated, so people were forced to vacate the nearby areas of the river to move to the safer place. Furthermore, according to the Assam daily flood reports (ADFRs) [15] from August

27 to September 5, 2021, the worst-hit areas of the Marigaon district were Mayong (B) and Bhuragaon (C) revenue blocks, whereas Laharighat (D) and Mikirbhetta (E) revenue blocks were the least affected. According to the report, the volunteer rescued many affected villagers from B and C revenue blocks of the Marigaon district. Hence, the rescue was done rigorously in this area. The distributions of the disaster damage across the distinct revenue blocks reported by the ADFRs is shown in Fig. 16(a). Using the proposed framework *SentimentMapper*, we also mapped the distributions of various emergency needs (rescue, EMS, food and shelter) of the affect people in the distinct revenue blocks of the Marigaon district (Fig. 16(b)–(e)). Comparing the outcome of the proposed framework with the damage proclaimed by ADFRs, we observed that the maximum and minimum rescue and EMS requests were collected from the worst-affected (revenue block B) and least-affected (revenue blocks D and E) regions, respectively. This implies that *SentimentMapper* correctly captures and labels the tweets to uncover the crucial situational information.

Thus, the results and findings ascertain that the proposed framework can collect and localize salient information from social media posts and provide insights to improve the disaster response and rescue plans.

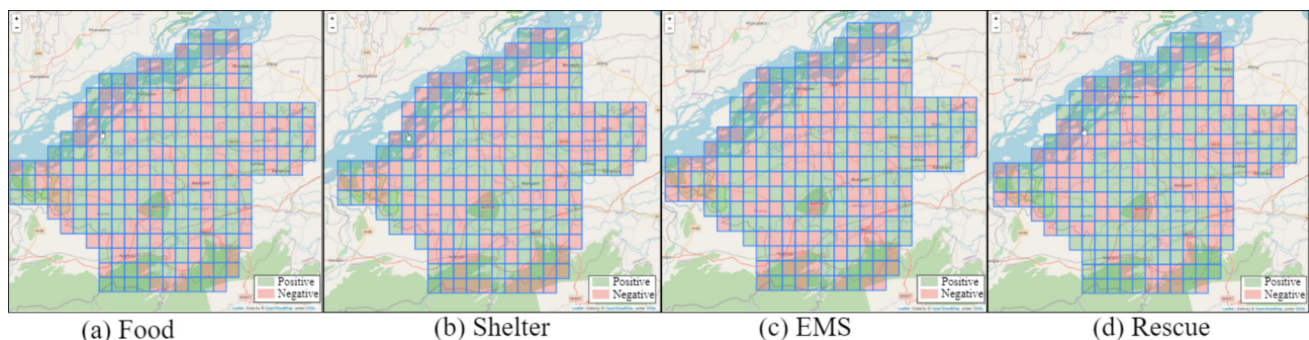


Fig. 15 Geographical distribution of the sentiments of people towards (a) Food, (b) Shelter, (c) EMS, and (d) Rescue

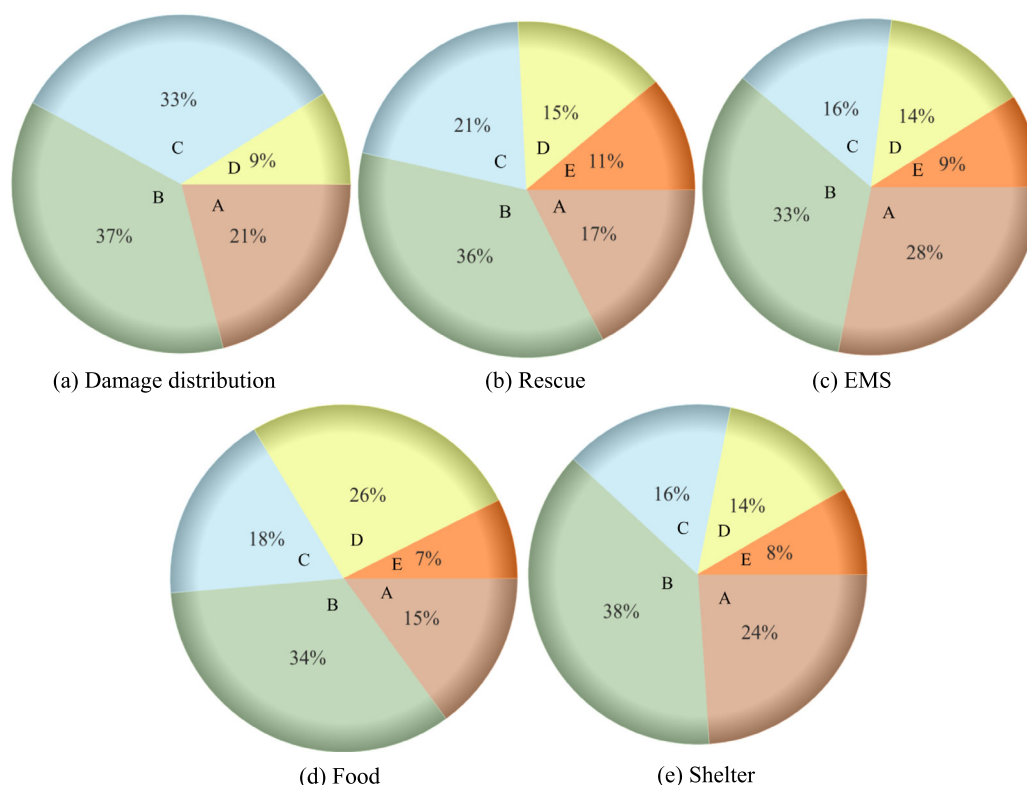


Fig. 16 Unfolding the disaster event. (a) The proportional distribution of the disaster damage reported by ASDMA in the different revenue blocks of the Marigaon district, (b)-(e) The distribution of victim claims

for various essential needs across the distinct revenue blocks of the Marigaon district (estimated by using *SentimentMapper*)

8 Limitations

The goal of the proposed work is to process and visualize the distribution of the sentiments of victims using social media data for effective disaster response. But there are some limitations of our study that are as follows:

- A notable limitation of the proposed framework is its vulnerability to potential attacks when social media platforms become inundated with false information. This scenario could arise if fake tweets are generated and disseminated by automated bots or fictitious accounts [16, 36]. To mitigate this issue, future efforts must address the challenges associated with implementing effective filtering mechanisms capable of detecting and flagging fake tweets and accounts.
- Another limitation of the proposed work is that it considers only English-language tweets as input. Although India is a country with a large number of vernacular languages, considering all or even a subset of them presents a significant challenge. The scarcity of resources for regional languages, such as Assamese, Bengali, Hindi, Malayalam, Oriya, Tamil, Telugu, and other regional languages, presents significant challenges [21]. These

challenges include the lack of available datasets and dictionaries specific to Indian languages, demanding novel research directions to address various NLP tasks using relevant corpora. As a result, further research efforts are necessary to explore innovative approaches and resource development to effectively tackle NLP tasks for these diverse linguistic contexts.

9 Conclusions and future work

This paper proposes an automated framework to analyze and map the sentiments of disaster-affected people towards different emergency needs like food, shelter, EMS and rescue during various disasters. The study also proposes a tweet-text independent approach to detect the location of the tweets posted on Twitter and discover the impact of the disaster events in different areas. The proposed framework can visualize the geographical distribution of the emergency needs and sentiments of the victims during and after a disaster, which previous studies have not done. Moreover, this framework can deliver better disaster response strategies and plan by drilling more helpful information from the social media data, like the percentage of positive and negative tweets in

case of different natural disasters, the percentage distributions of tweets regarding various crucial requirements, public opinion toward various identified emergency needs, and the geographical distribution of the emergency needs for the affected people. The case study findings suggest that the proposed framework can effectively fetch the spatial distribution of the public sentiments about emergency needs and provide a lucid overview of the disaster-affected area to the rescue personnel. Additionally, the *SentimentMapper* allows us to develop basic statistics like the distribution of the emergency needs of the people and statistics of the most vulnerable areas in the disaster hit area. Such facts can be incredibly helpful for policymakers to understand the disaster situation and to manage the disaster relief and response resources required for the affected community. Therefore, the proposed framework will provide a micro-level understanding to the disaster managers regarding the crisis and assist them in preparing a more efficient and effective disaster response plan.

In the future, this study can be extended to incorporate additional data sources, such as images shared on social media platforms, satellite imagery, and drone-captured images, to gain a more comprehensive understanding of disaster-affected areas. Additionally, the current work focuses exclusively on English-language tweets, whereas during crises, users frequently post in their regional languages. To enhance the adaptability of the framework to diverse disaster scenarios, the next phase of this research will focus on developing a multilingual model capable of processing tweets in multiple languages.

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Data Availability Data used in the methodology proposed in this manuscript will be available on request.

Declarations

Competing Interests: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Compliance with Ethical Standards: The authors of this article did not conduct any studies involving human participants.

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